首都师范大学
CAPITAL NORMAL UNIVERSITY

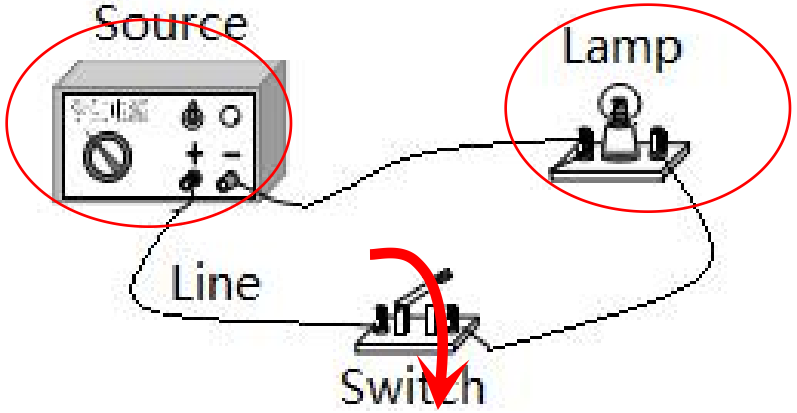
电路分析

电路的基本概念及定律

主讲人：朱 虹

一 电路（circuits ）

电路　电流的闭合路径

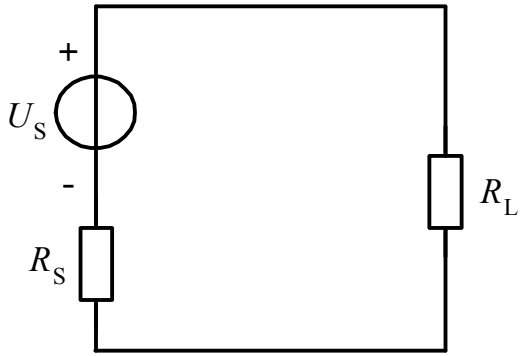


电路
- 电源（ source ）
  - 电力源
  - 信号源
- 元件/负载（element/load)
- 导线（line）、开关（switch)

功能

①、实现能量的转换 ：电能 ⟷ 其他能

②、实现信号的处理和传输



电路模型　理想元件、导线　理想电路

二 → 电流（current）

1 电流强度

电荷的定向运动形成电流

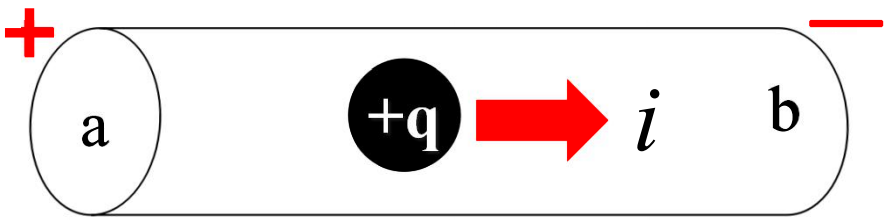
电流强度 $i(t)=\lim_{\Delta t\to 0}\frac{\Delta q}{\Delta t}=\frac{\mathrm{d}q(t)}{\mathrm{d}t}$

库仑 /C

秒 /s

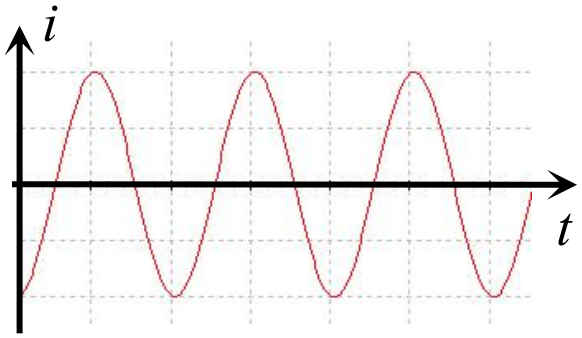
安培/A

kA/ mA/ uA



$i(t)=\mathrm{C}$ 直流电流/DC $i(t)\to I$

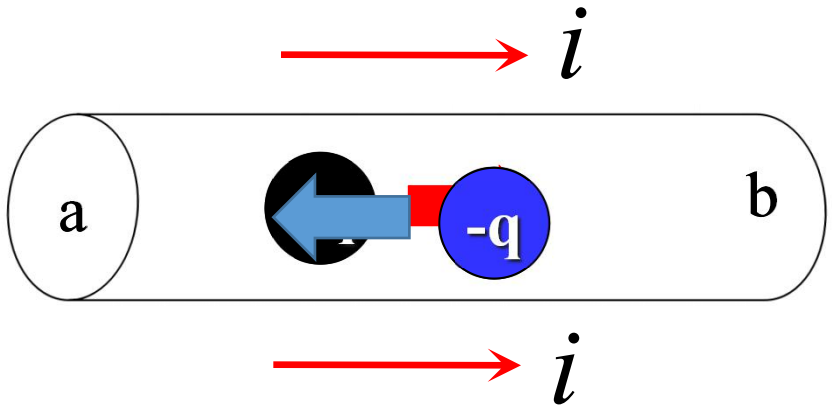
$i(t)=I_{\mathrm{m}}\sin(\omega t+\varphi_{\mathrm{i}})$ 正弦交流电流/AC $i(t)\to i$



二 → 电流（current）

2 电流的参考方向（reference direction）

电流方向　**规定正电荷的运动方向**



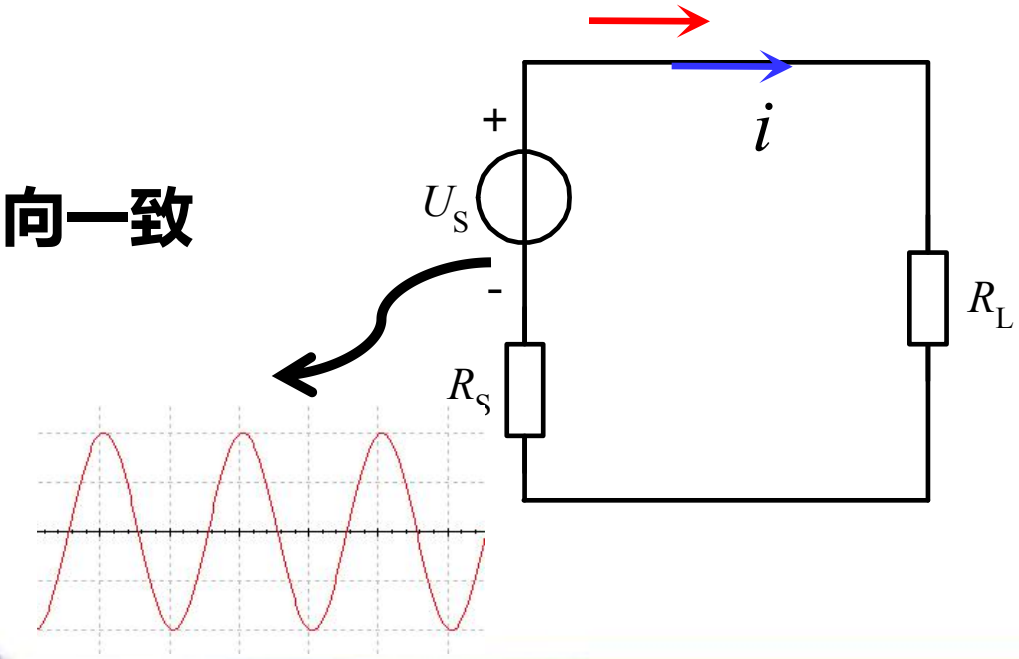
实际方向（real direction）

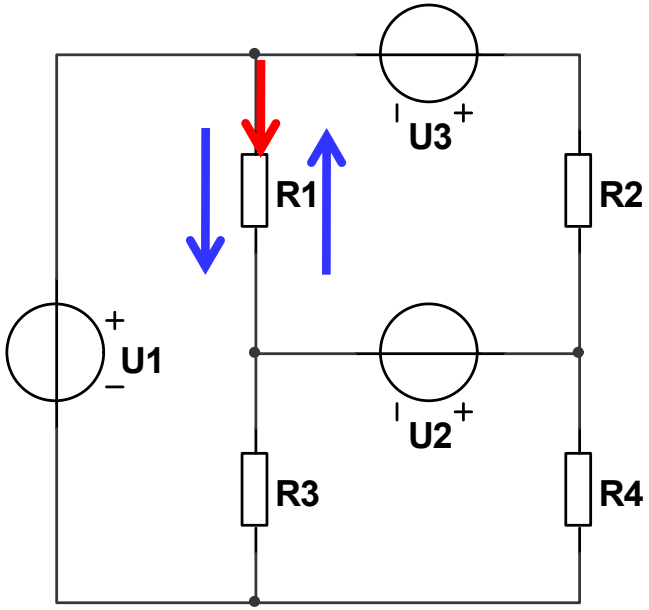
参考方向（reference direction）

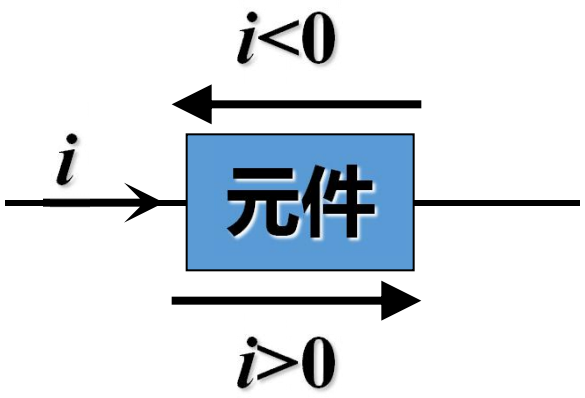
假设一个方向，用+、-表示

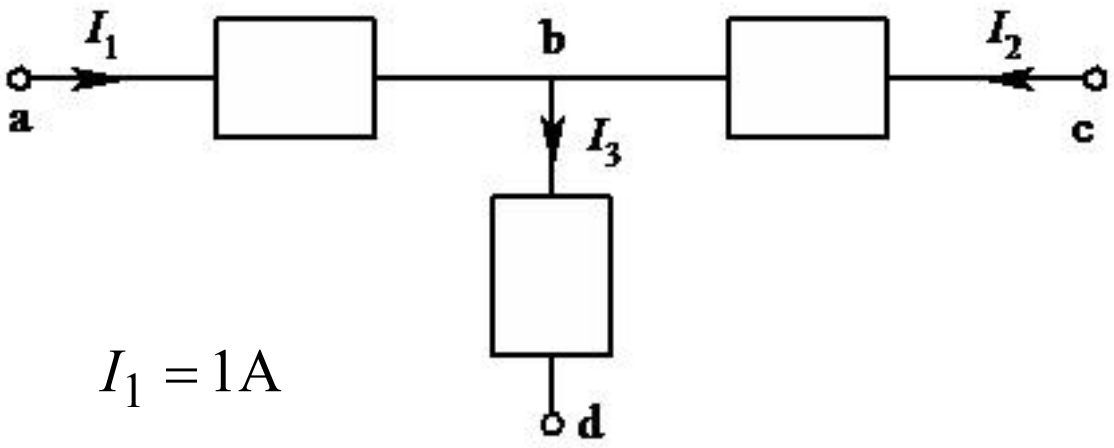
若 *i* 为＋，

参考方向与实际方向一致









$$I_1 = 1\text{A}$$

$$I_2 = -2\text{A}$$

$$I_3 = -1\text{A}$$

以参考方向作为电流方向，不再考虑实际方向

三 → 电位与电压（potential & voltage）

1 电位（potential）

$$u_a = \lim_{\Delta q \to 0} \frac{\Delta w}{\Delta q} = \frac{dw}{dq}$$

能量/J

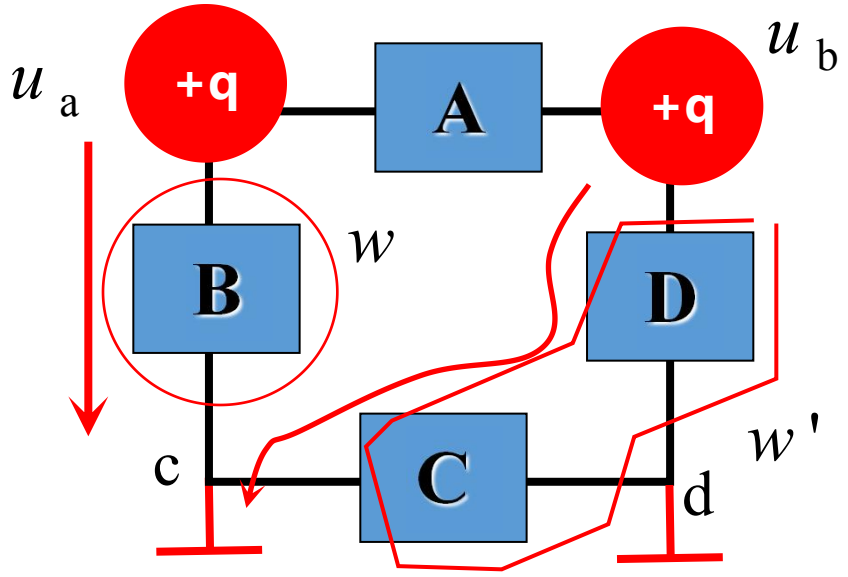
电荷量/C

电位 V

Δw

吸收能量 $\Delta w > 0$ $\quad u_a > 0 \quad u_a > u_c \quad u_a : +$ 元件

释放能量 $\Delta w < 0$ $\quad u_a < 0 \quad u_a < u_c \quad u_a : -$ 电源



参考点 (GND)

$u_d = 0$

$u_c = 0$

电位相对性 参考点不同，各点的电位不同

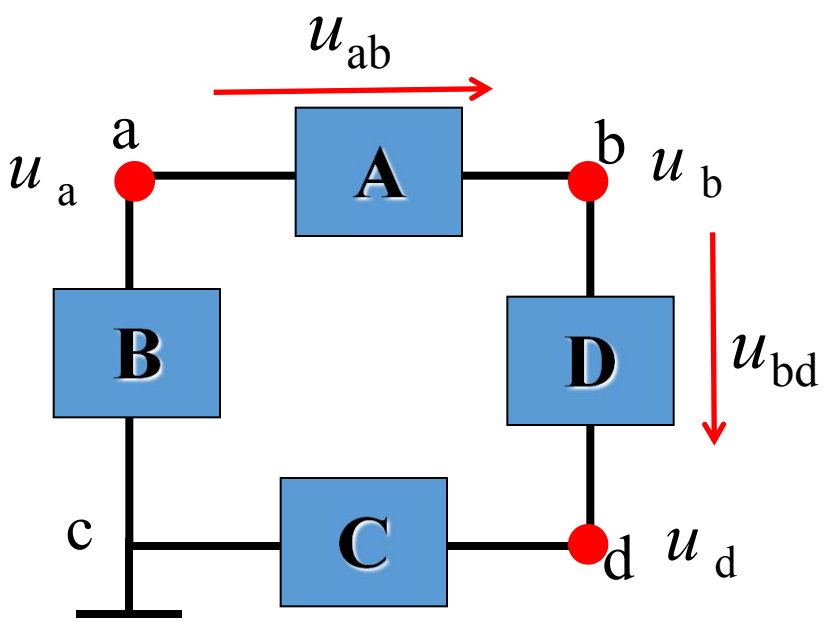
电子电路中，一般取电源的负极为参考点

三 → 电位与电压（potential & voltage）

2 电压（voltage）

电压： 两点间的电位差

$$u_{ab} = u_a - u_b \qquad u_{bd} = u_b - u_d$$

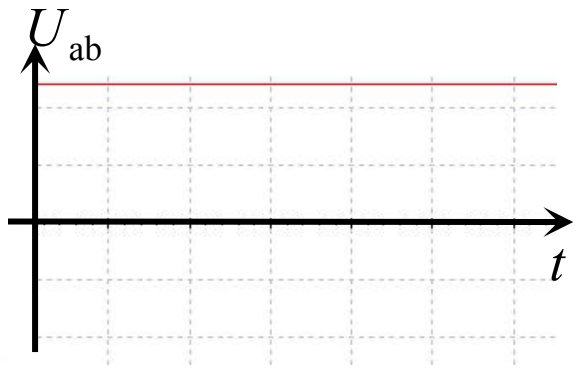


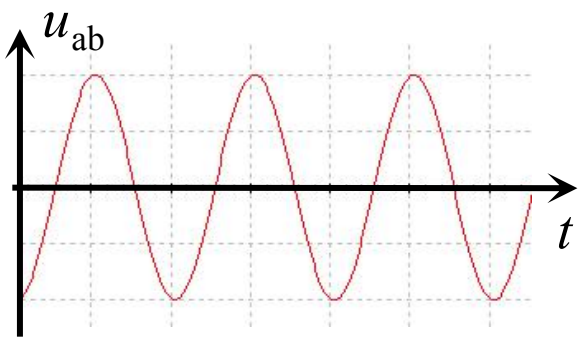
电压的绝对性 **电压与参考点无关**

直流电压（DC） $u_{ab} = C$ U_{ab}

正弦交流电压（AC） $u_{ab}(t) = U_m \sin(\omega t + \varphi_u)$

u_{ab}



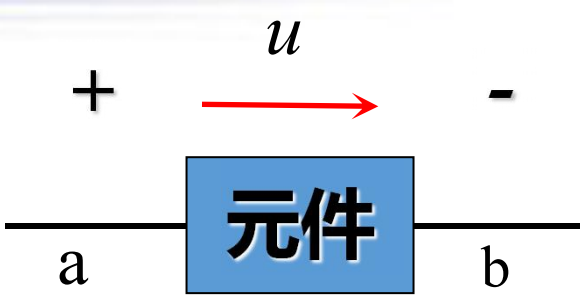


三 → 电位与电压（potential & voltage）

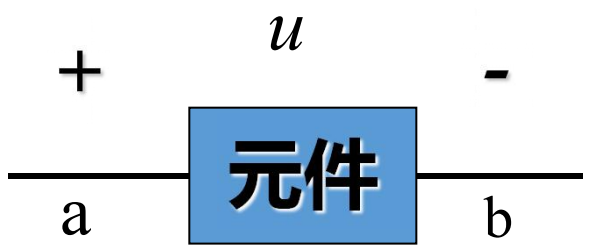
3 电压的参考方向（reference direction）

电压的方向/ 极性　高电位点标＋，低电位点标－

$$u_{ab} = u_a - u_b$$



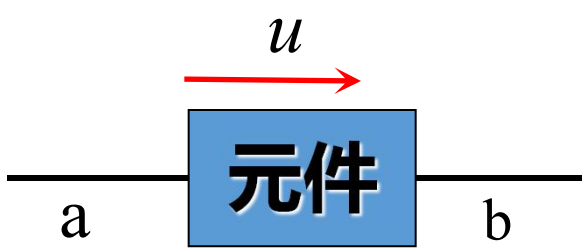
电压的参考方向

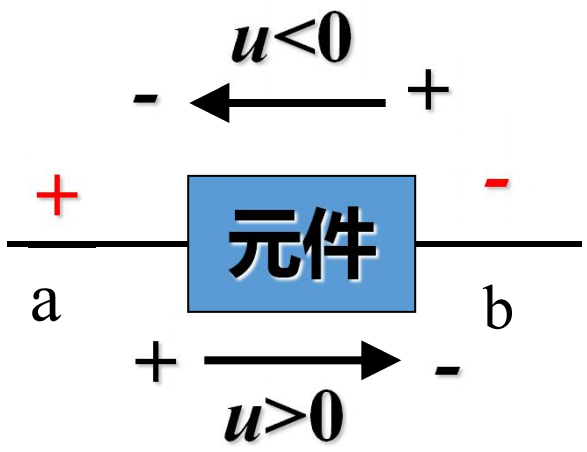


$$u_{ab} = u_a - u_b$$

a点为＋，b点为－

$$u_{ba} = u_b - u_a$$



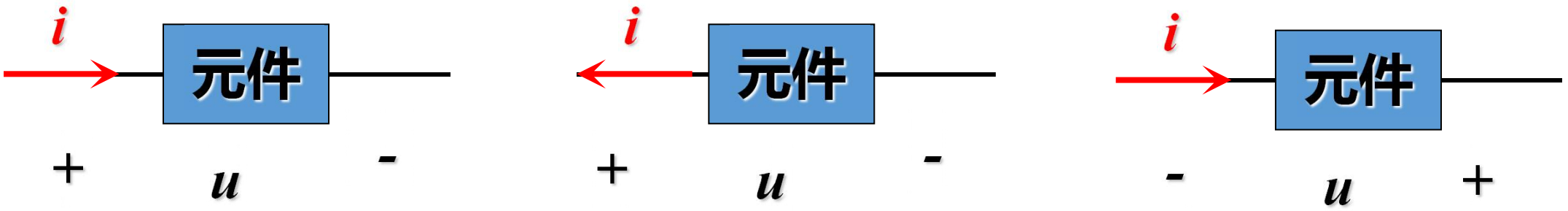


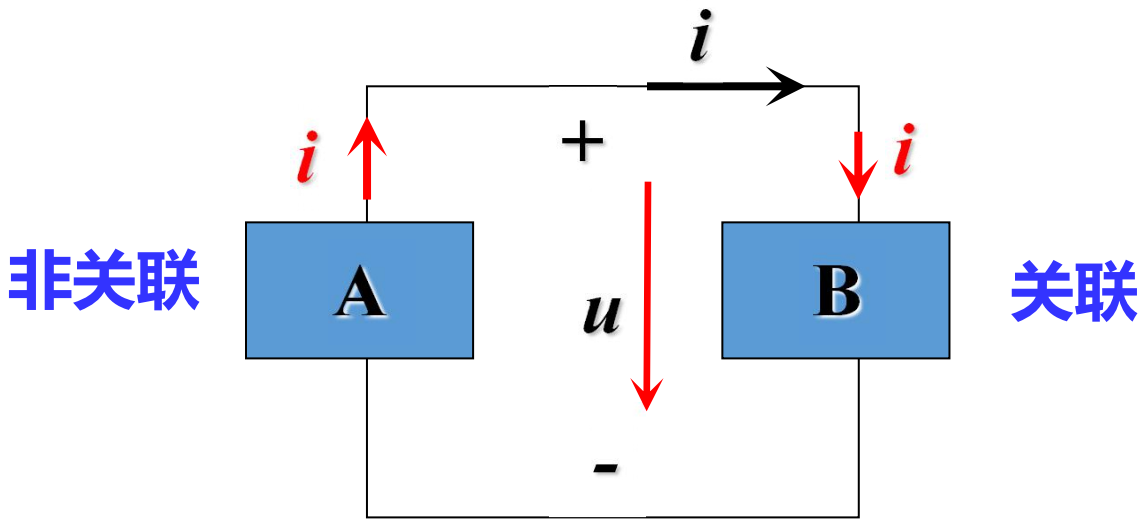
三 → 电位与电压（potential & voltage）

4 关联参考方向
（associated reference directions）

关联参考方向

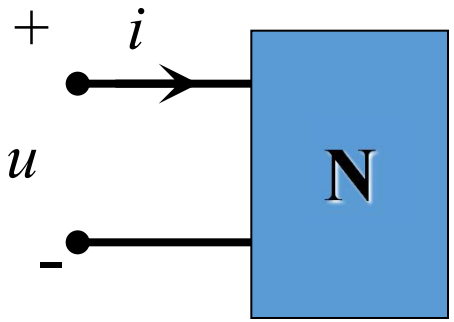
非关联参考方向





四 → 电功率与能量（power & energy）

功率



关联参考方向下：

$$p = \lim_{\Delta t \to 0} \frac{\Delta w}{\Delta t} = \frac{\mathrm{d}w}{\mathrm{d}t} = \overset{u}{\frac{\mathrm{d}w}{\mathrm{d}q}} \cdot \overset{i}{\frac{\mathrm{d}q}{\mathrm{d}t}} = u\,i$$

瓦特/W

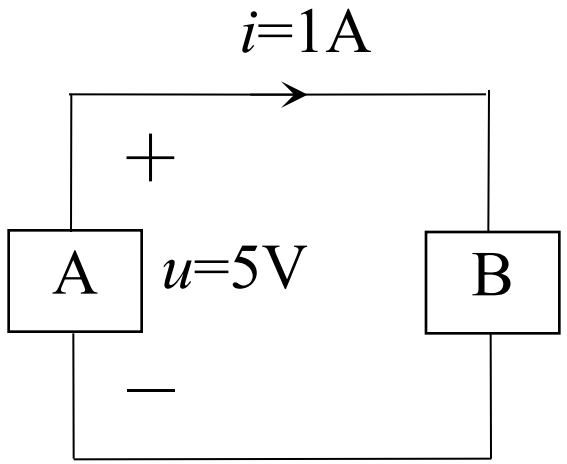
非关联参考方向下： $p = -u\,i$

***p*>0：吸收能量** ***p*<0：释放能量**

能量 $w = \int_{t_0}^{t} p\mathrm{d}t = \int_{t_0}^{t} u\,i\mathrm{d}t$ **焦耳/J** **度（kWh）**

直流电压/电流 $W = UI(t - t_0)$

例　求A、B两个元件的功率各为多少？并说明它们分别是吸收还是放出功率。



解：　A：非关联参考方向

$$p_A = -ui = -5\times 1 = -5(\mathrm{W})$$

A释放功率5W

B：关联参考方向

$$p_B = ui = 5\times 1 = 5(\mathrm{W})$$

B吸收功率5W

$$p_A + p_B = 0$$

功率守恒性

五 → 基尔霍夫定律（Kirchhoff's Law）

1 支路、节点、回路、网孔
（branch, node, loop & mesh ）

1、支路（branch）

单个或多个元件串联 $b=6$

2、节点（node）

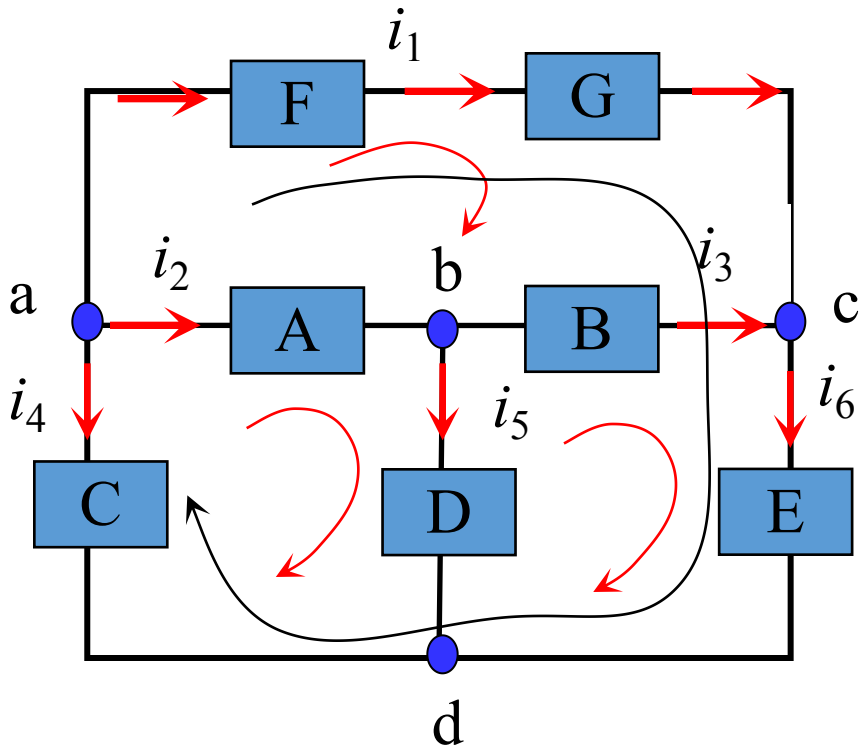
三条以上支路的连接点 $n=4$

3、回路（loop）

闭合路径

4、网孔（mesh）

内部不再包含支路的回路 $m=3$



$$m=b-n+1$$

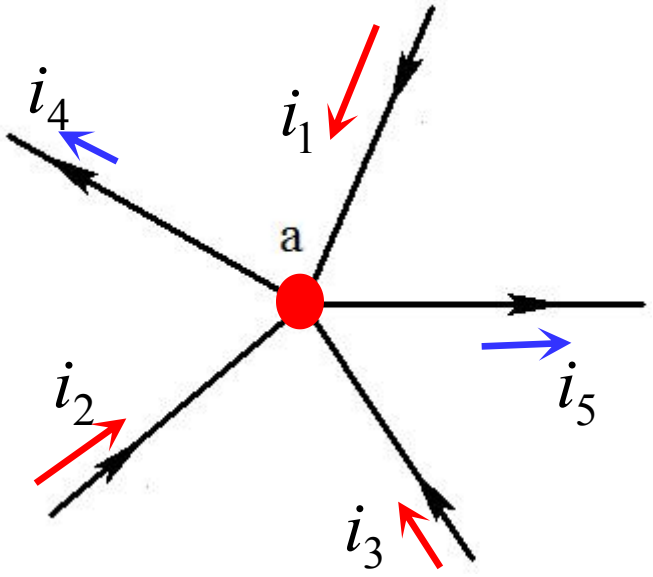
五 → 基尔霍夫定律（Kirchhoff's Law）

2 基尔霍夫电流定律（KCL)
（Kirchhoff's Current Law ）

2、基尔霍夫电流定律（KCL）

任意电路、任意节点、任意时刻

流入或流出该节点的电流之和恒等于0



设流入为＋，流出为－

$$\sum_{k=1}^{b} i_k = 0$$

$$i_1 + i_2 + i_3 - i_4 - i_5 = 0$$

$$i_1 + i_2 + i_3 = i_4 + i_5$$

$$\sum i_{流入} = \sum i_{流出}$$

线性电路
非线性电路

与元件无关

直流/交流
周期/非周期

电路的拓扑关系的约束

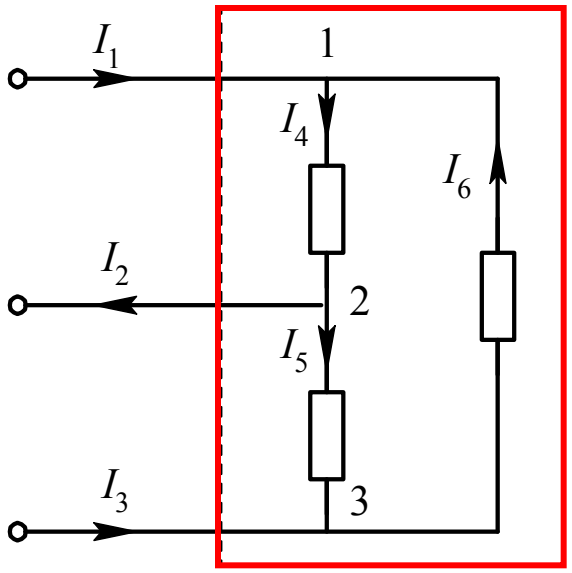
本质：电荷守恒定律

节点1： $i_1 + i_6 = i_4$

节点2： $i_4 = i_2 + i_5$

节点3： $i_3 + i_5 = i_6$

$$i_1 + i_3 = i_2$$



广义节点/ 超节点

KCL的实质：电荷守恒

五 → 基尔霍夫定律（Kirchhoff's Law）

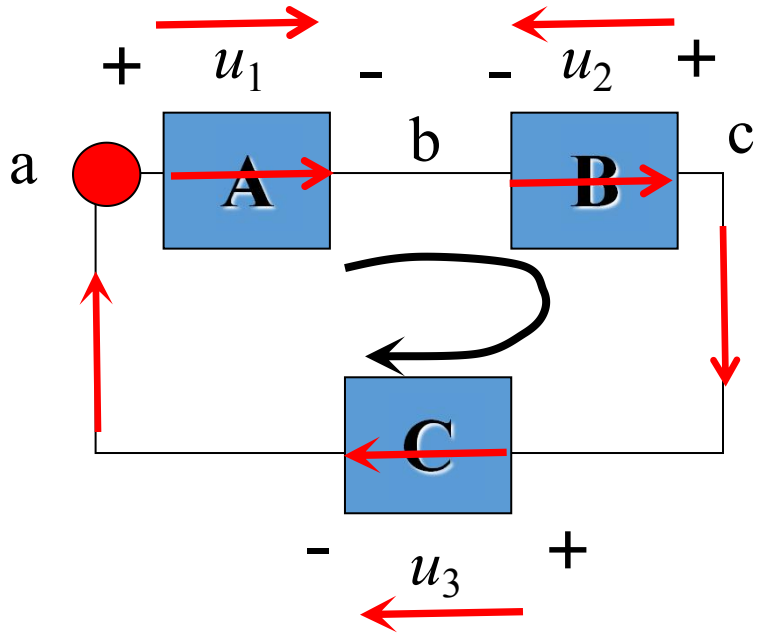
3 基尔霍夫电压定律（KVL)
（Kirchhoff's Voltage Law ）

3、基尔霍夫电压定律（KVL）

任意电路、任意回路、任意时刻

沿回路方向，所有元件的电压降之和恒等于0

选取回路方向，设与回路同向为正，反向为负



$$u_1 - u_2 + u_3 = 0 \qquad \sum_{k=1}^{m} u_k = 0$$

$$u_1 + u_3 = u_2 \qquad \sum u_{降} = \sum u_{升}$$

$$= u_a - u_a = 0$$

$$u_{aa} = u_{ab} + u_{bc} + u_{ca}$$

$$= u_1 - u_2 + u_3$$

线性电路
非线性电路

与元件无关

直流/交流
周期/非周期

电路的拓扑关系的约束

五 → 基尔霍夫定律（Kirchhoff's Law）

4 基尔霍夫定律例题

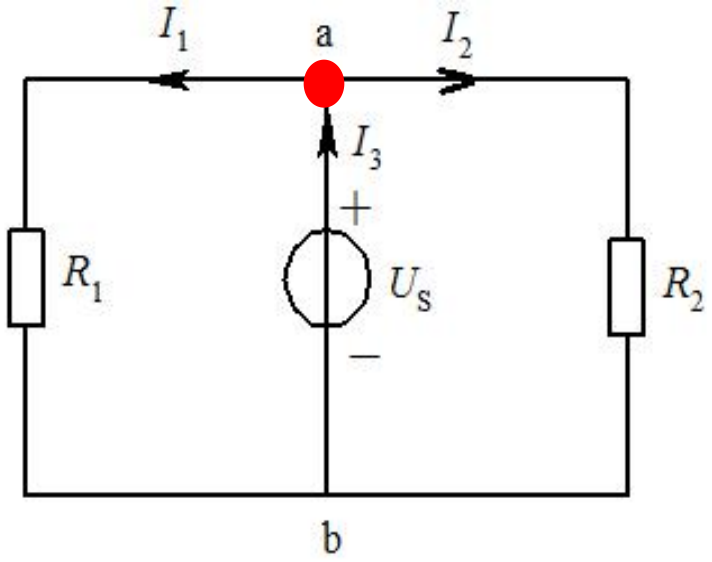
例1　已知：R_1=2Ω ，R_2=5Ω，U_S=10V。求各支路电流。

解：

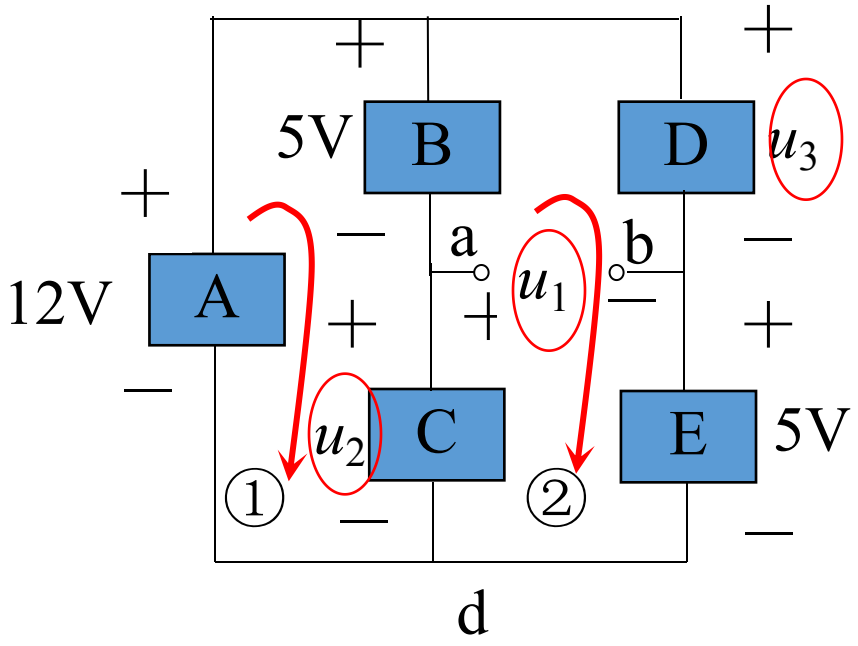
$$I_1 = \frac{U_S}{R_1} = \frac{10}{2} = 5\ (\mathrm{A})$$

$$I_2 = \frac{U_S}{R_2} = \frac{10}{5} = 2\ (\mathrm{A})$$

$$I_3 = I_1 + I_2 = 5 + 2 = 7\ (\mathrm{A})$$



例2　电路如图，求 u_1, u_2, u_3



解：　回路①：　$5+u_2=12$

$u_2=7\text{V}$

回路②：　$u_3+5=5+u_2$

$u_3=u_2=7\text{V}$

取d点为参考点，则　$u_d=0$

$u_a=u_2=7\text{V}$　　$u_b=5\text{V}$

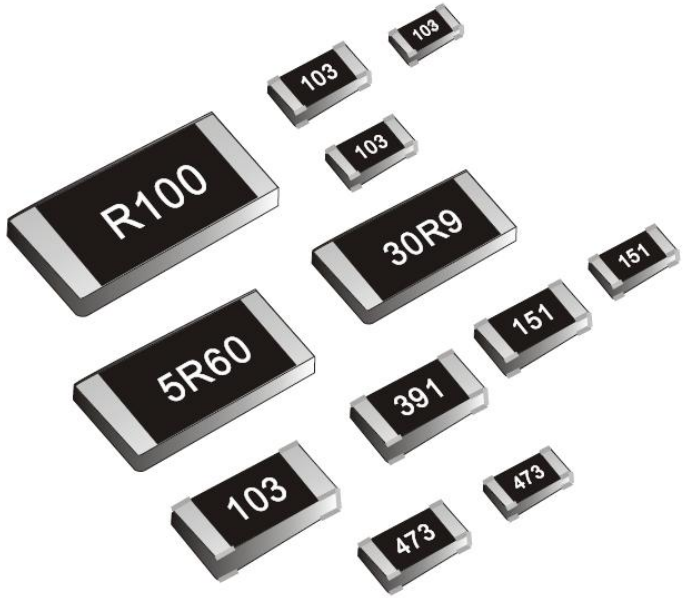
$u_1=u_a-u_b=7-5=2\text{V}$

六 → 电阻元件（resistor）

1 电阻元件（resistor ）

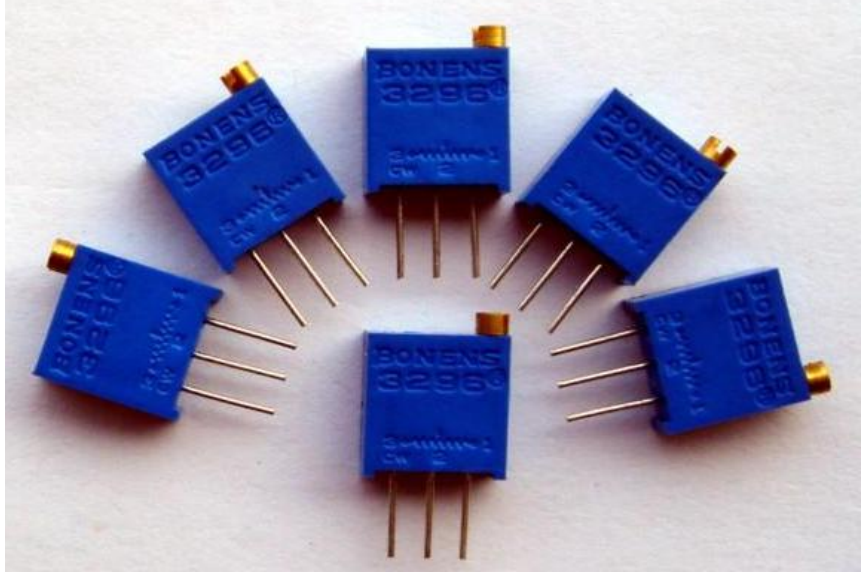
实际中的电阻器

longde

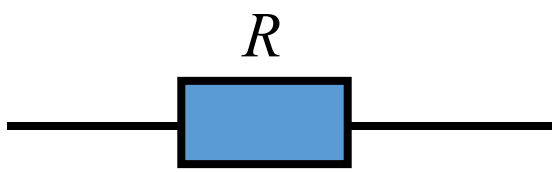

R100
103
103
103
30R9
5R60
151
151
391
103
473
473

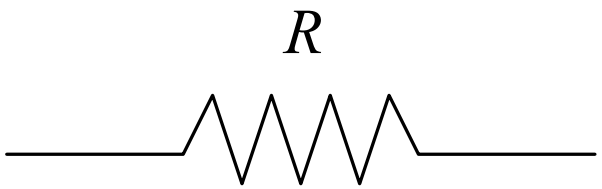
MUSIC

可变电阻器与电位器



电阻元件的符号及参数





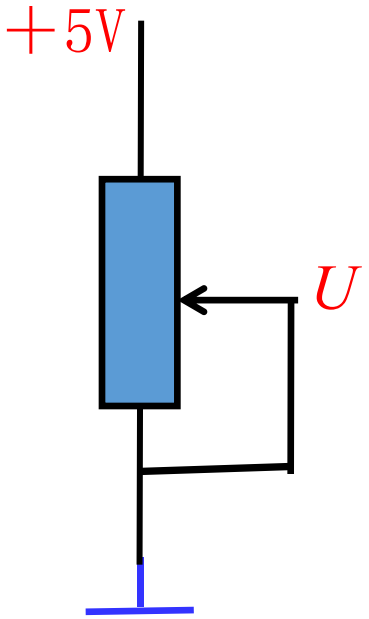
电阻元件的参数：

- **电阻值（resistance）**

  51Ω，10kΩ，1MΩ…

- **额定功率**

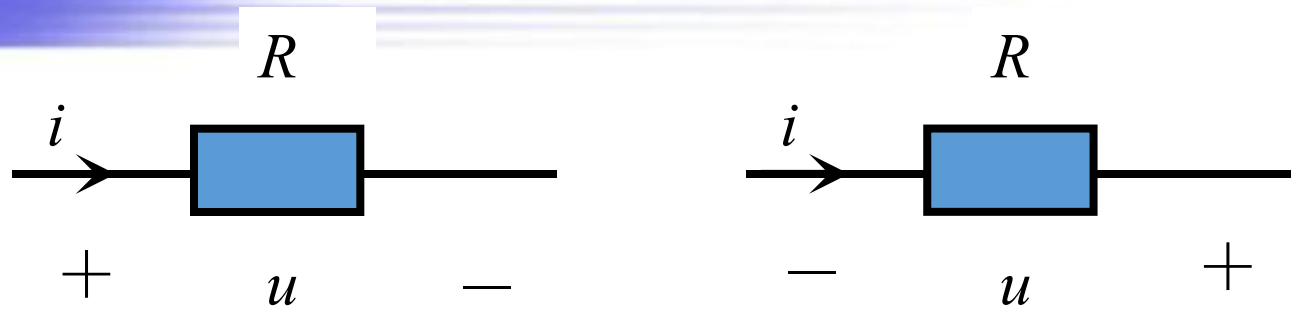
  1/8 W，1/4 W，2W…



六 → 电阻元件（resistor）

2 电阻元件的伏安关系（VAR）（volt-ampere relationship）

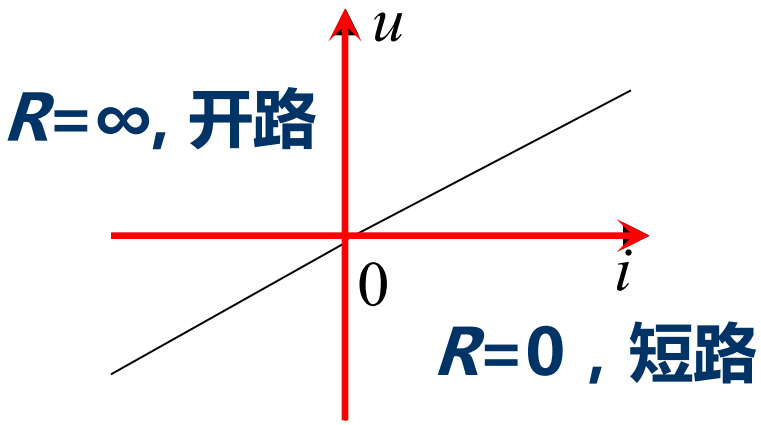
欧姆定律 (Ohm's Law)



关联方向： $u = Ri$

欧姆/Ω

非关联方向： $u = -Ri$

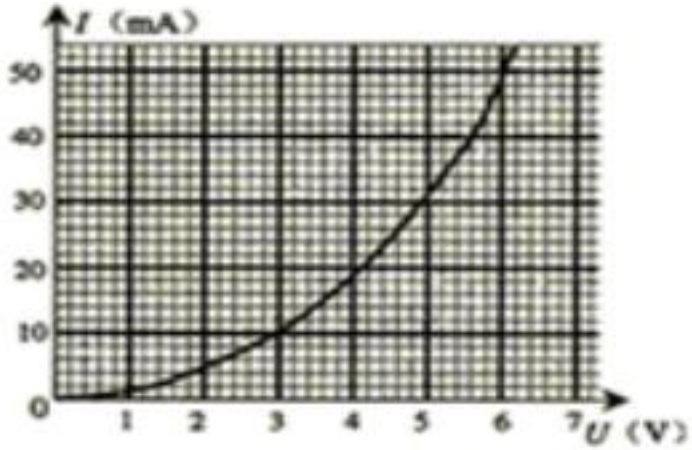


线性电阻　电阻*R*是定值　VAR是线性函数

非线性电阻　电阻*R*不是定值　如：热敏电阻

$$u = Ri \qquad i = \frac{u}{R} = G \cdot u$$

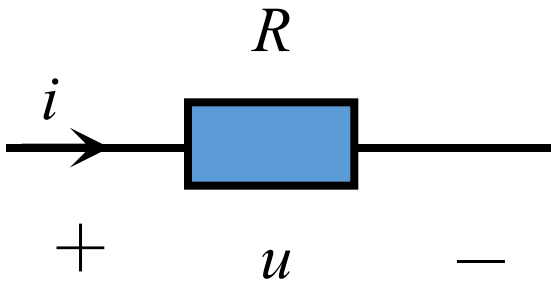
电导（conductance） $G = \dfrac{1}{R}$　1/ Ω　西门子/S



六 → 电阻元件（resistor）

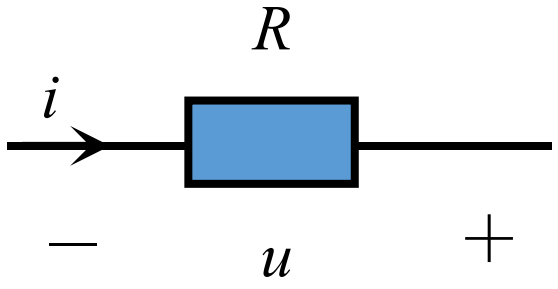
3 电阻元件的功率

$$p = ui = Ri^2 = \frac{u^2}{R} = Gu^2 \geq 0$$



$$p = -ui = Ri^2 = \frac{u^2}{R} \geq 0$$

$$u = -Ri$$

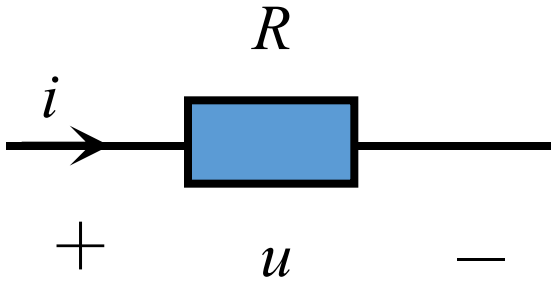


$p_R \geq 0$ **吸收电能** **耗能元件**

额定功率： $1\text{W}, \frac{1}{4}\text{W}, \frac{1}{8}\text{W}, \frac{1}{16}\text{W}$

例1　已知： $u = 10\sin 2t\ \mathrm{V}\quad i = 2\sin 2t\ \mathrm{A}$ **求*R*。在5秒内消耗的电能是多少？**

解： $R = \dfrac{u}{i} = \dfrac{10\sin 2t}{2\sin 2t} = 5(\Omega)$



$$p = ui = 10\sin 2t \cdot 2\sin 2t = 20\sin^2 2t\,(\mathrm{W})$$

$$W = \int_0^5 p\,\mathrm{dt} = \int_0^5 20\sin^2 2t\,\mathrm{dt} = 10\int_0^5 (1-\cos 4t)\,\mathrm{dt}$$

$$= 10\left(t - \frac{1}{4}\sin 4t\right)\Big|_0^5 = 49.14\,(\mathrm{J})$$

例2　R=4Ω，额定功率为0.25W处于直流电路中，问使用时最大的限定电流是多少？

解：　$P = RI^2 \le 0.25\ \mathrm{W}$

$$I = \sqrt{\frac{P}{R}} \le \sqrt{\frac{0.25}{4}} = \frac{1}{4}(\mathrm{A}) = 250(\mathrm{mA})$$

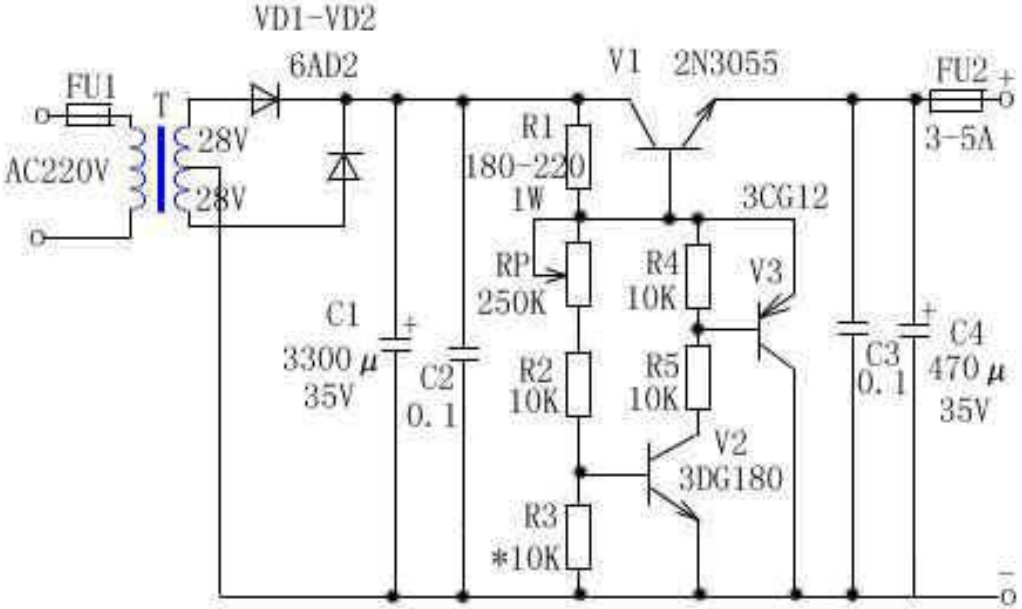
七 → 独立电源（independent source）

1 电源与信号源
（power source & signal source）

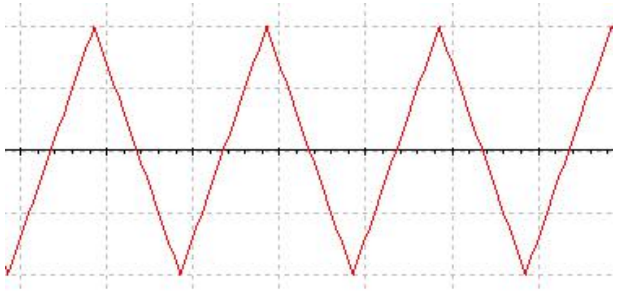
电源（Power sources）

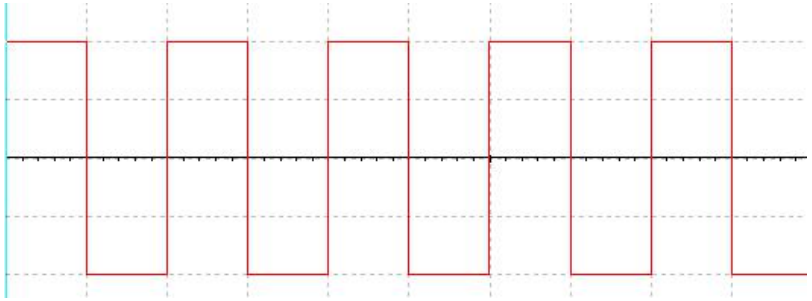
电源（Power sources）

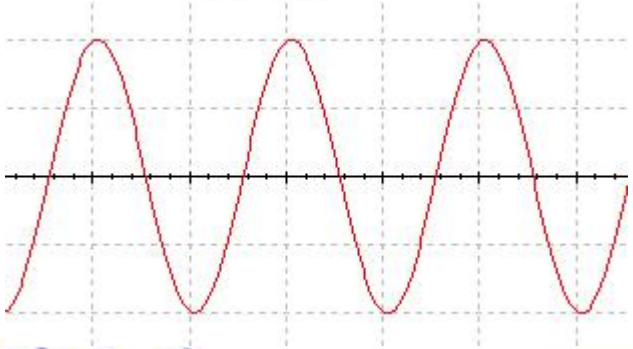
直流稳压电源



信号源（Signal sources）





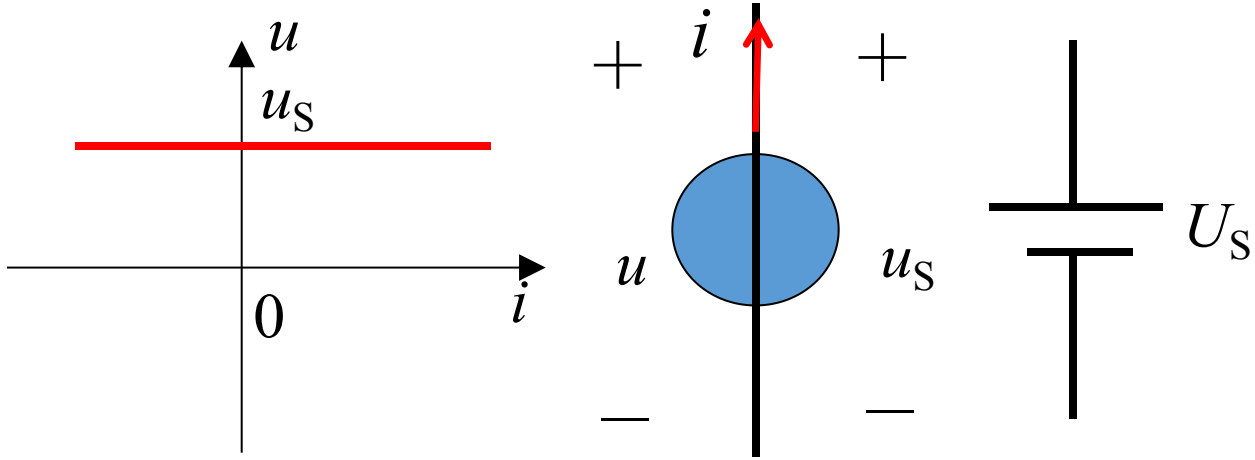


七 → 独立电源（independent source）

2 理想电压源（ideal voltage source ）

电压源　　恒压源

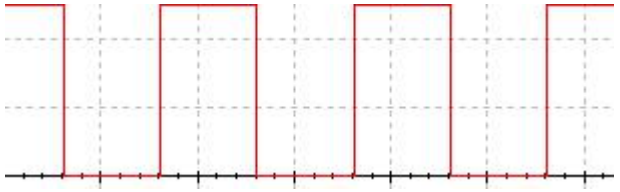
伏安特性　$u = u_S$　$u_S(t)$

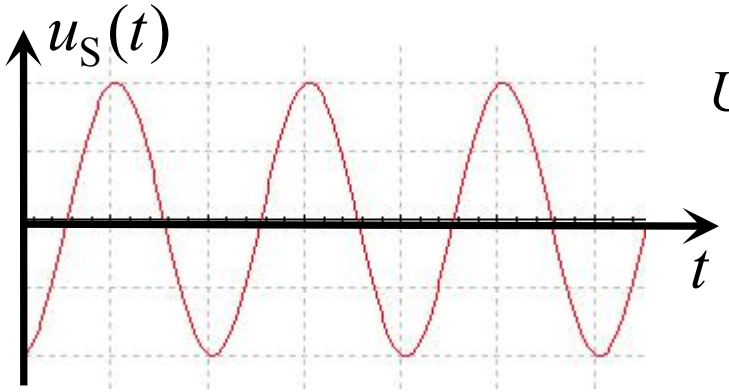


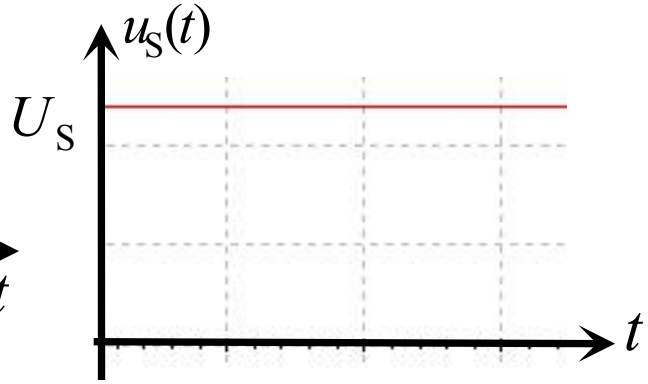
交流电压源(AC)　$u_S(t) = U_m \sin(\omega t + \varphi_u)$

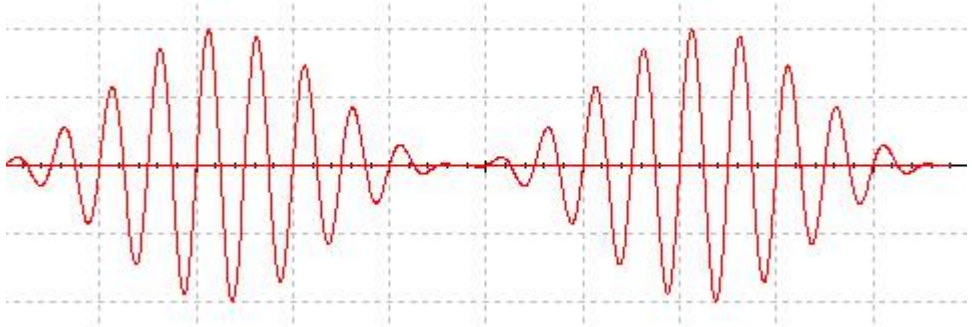
直流电压源(DC)　$u_S(t) = C$　U_S

信号源

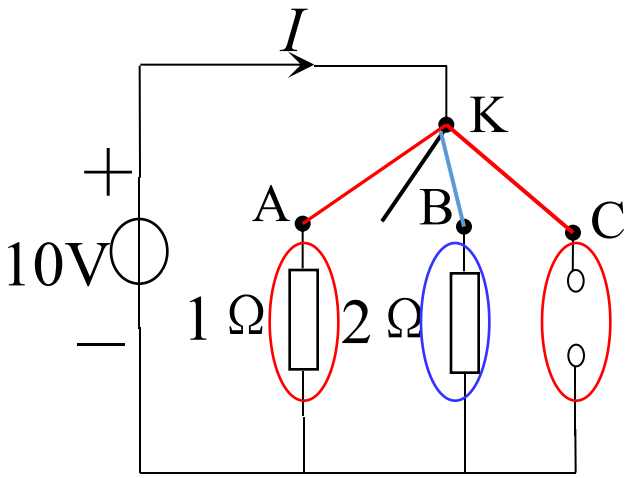








例　如图，求开关分别在A、B、C时的电流 *I*。



解：A点　$I=\dfrac{10}{1}=10(\mathrm{A})$

B点　$I'=\dfrac{10}{2}=5(\mathrm{A})$

C点　$I''=0$

端电压不随电流的变化而变化　端电流由外电路决定

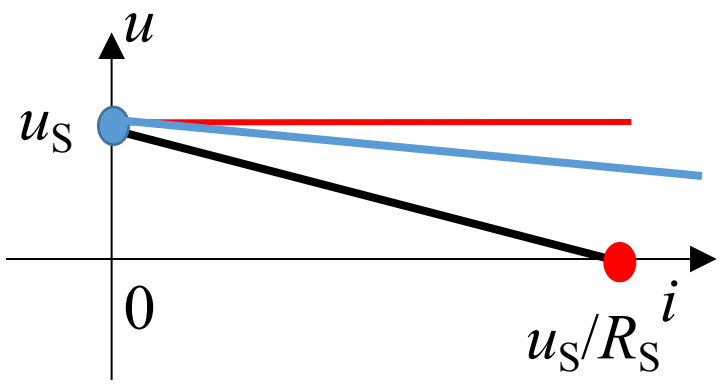
七 → 独立电源（independent source）

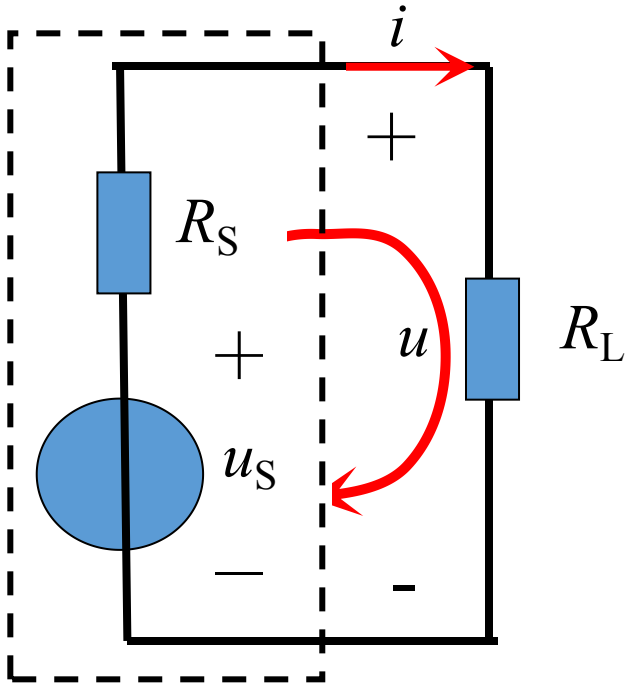
3 实际电压源（practical voltage source ）

伏安特性 $u = u_s - R_s i$

$i=0$，空载 $u = u_S$

$u=0$，短路 $i = \frac{u_S}{R_S}$ ↑ ↓





R_S 越小，电压源特性越好

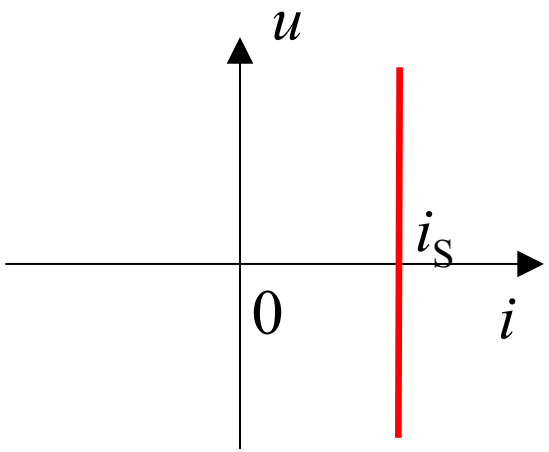
电压源两端不允许短路

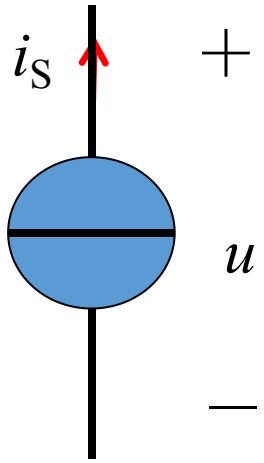
七 → 独立电源（independent source）

4 理想电流源（ideal current source ）

电流源　　恒流源

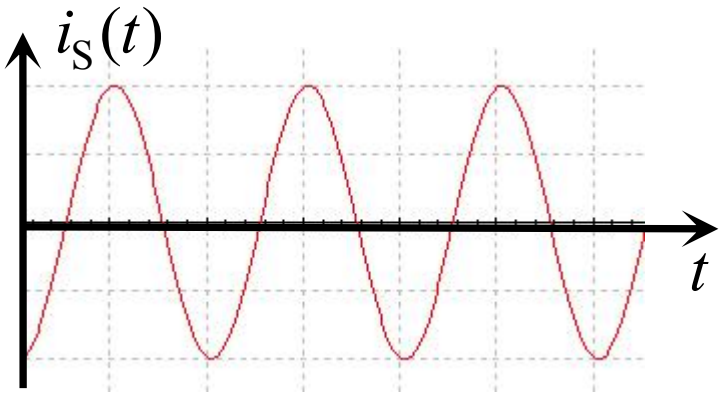
伏安特性　$i = i_S$　$i_S(t)$

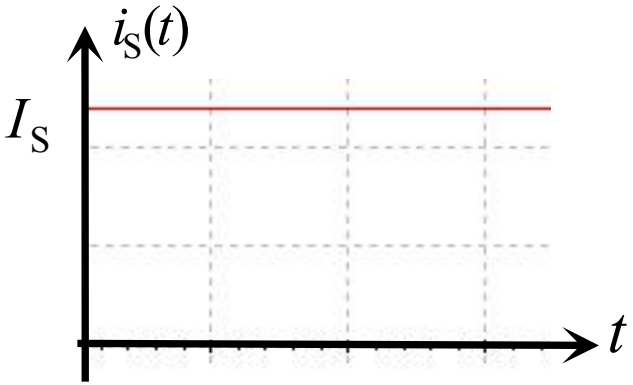




交流电流源(AC)　$i_S(t) = I_m \sin(\omega t + \varphi_i)$

直流电流源(DC)　$i_S(t) = C$　I_S

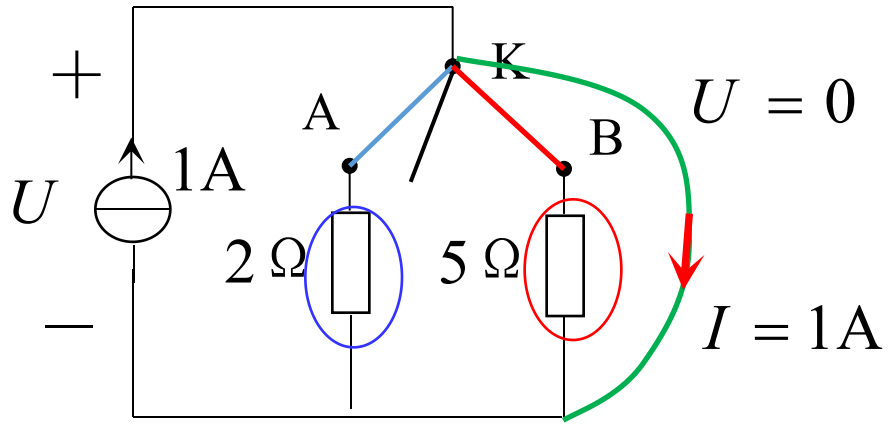




例　如图，求开关分别在A、B点时的电压 *U*。

解：A点　$U = 1 \times 2 = 2\ (\mathrm{V})$

B点　$U = 1 \times 5 = 5\ (\mathrm{V})$



端电流不随电压的变化而变化　端电压由外电路决定

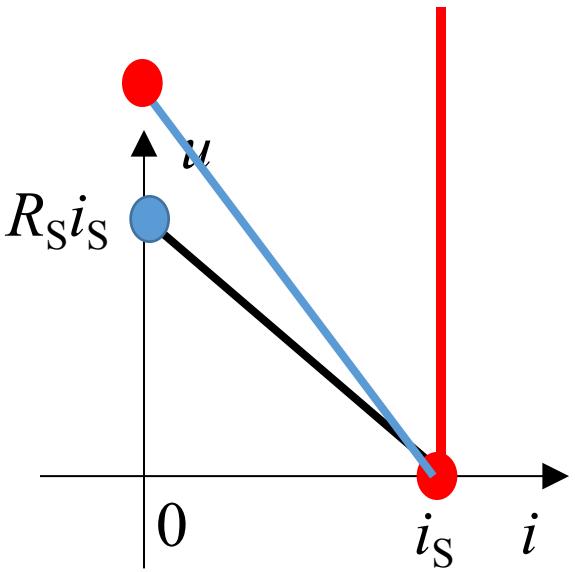
七 → 独立电源（independent source）

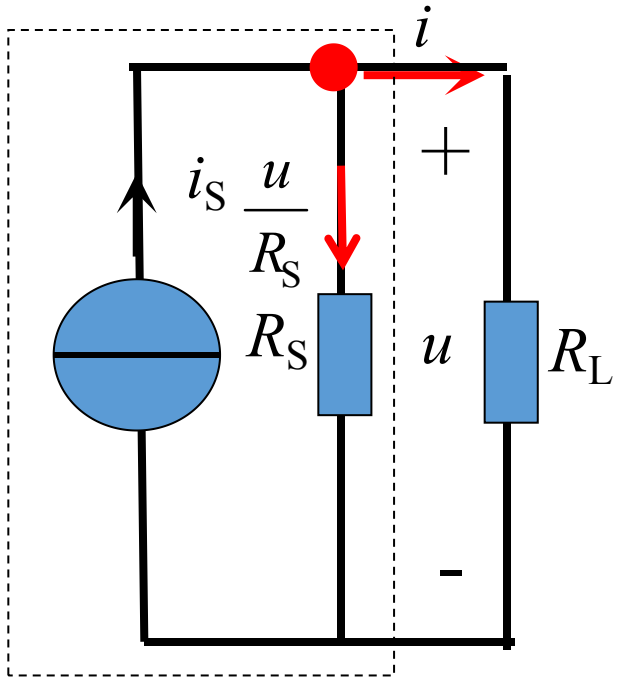
5 实际电流源（practical current source ）

伏安特性 $i = i_S - \dfrac{u}{R_S}$

$u = 0$，短路 $i = i_S$

$i = 0$，开路 $u = R_S i_S$





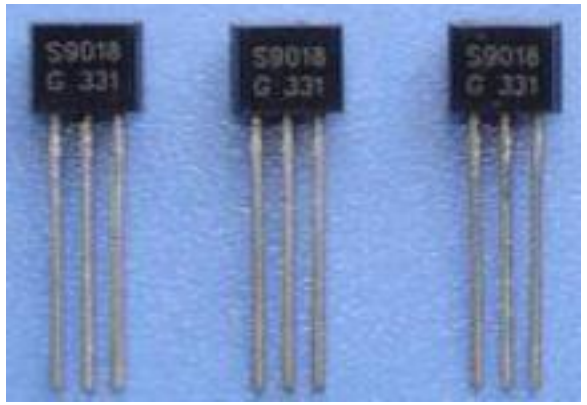
R_S 越大，电流源特性越好

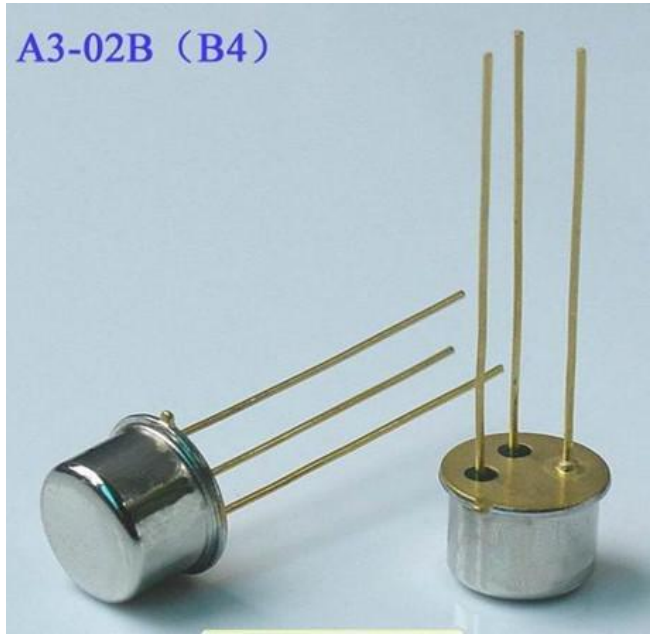
电流源两端不允许开路

八 → 受控源（controlled source）

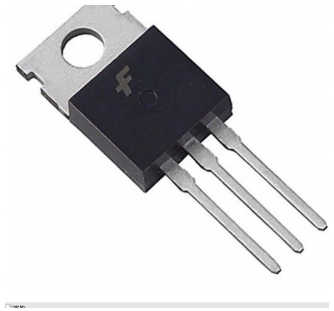
1 受控源（controlled source）

晶体管（Transistor）





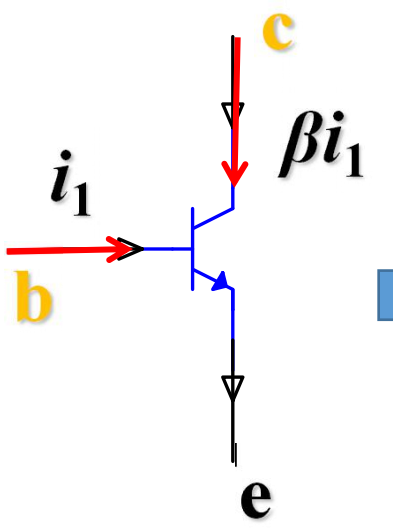
场效应管/MOSFET

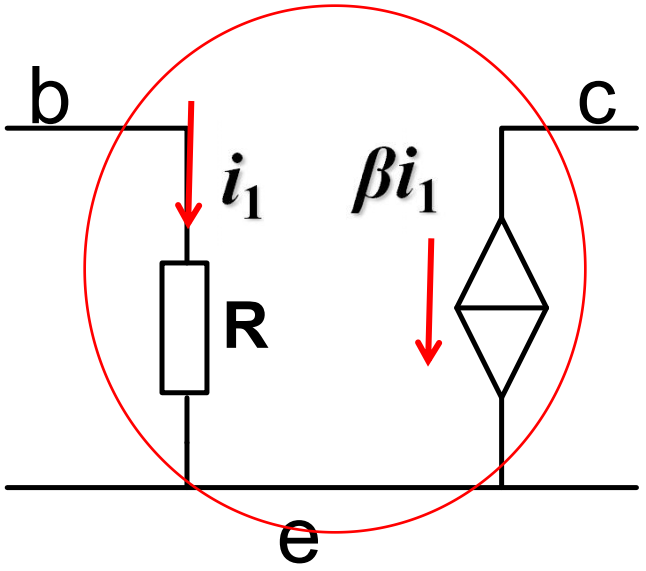


运算放大器
（Operation amplifer）

双极型晶体管

电流控制电流源（CCCS）



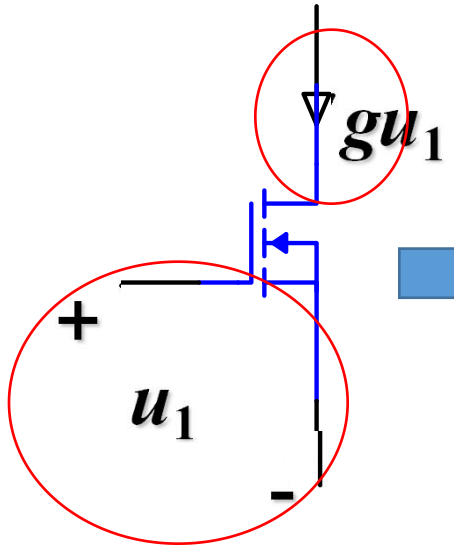


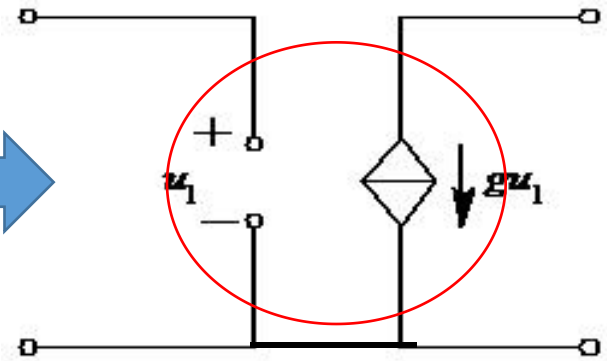
β 为电流放大倍数

β 为控制系数，无量纲

场效应管/MOSFET

电压控制电流源（VCCS）



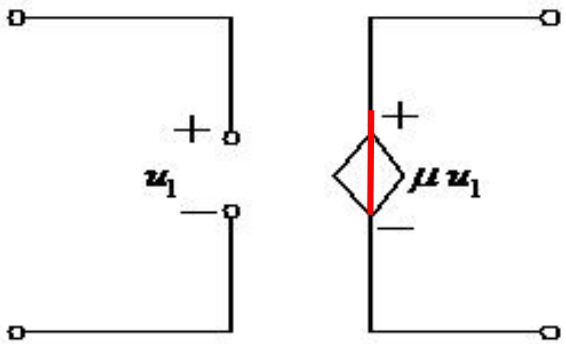


g为跨导

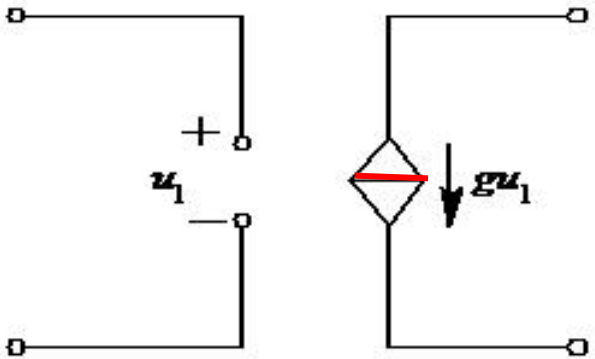
g为控制系数， S

受控源类型

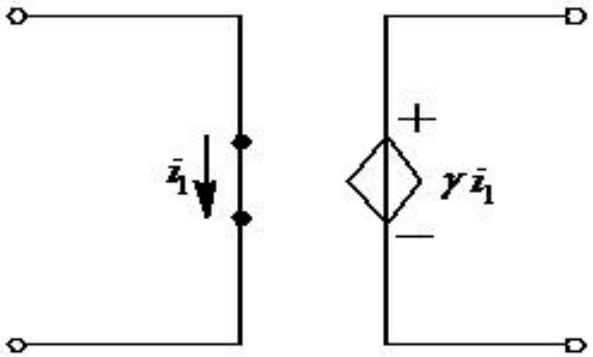
电压控制电压源VCVS



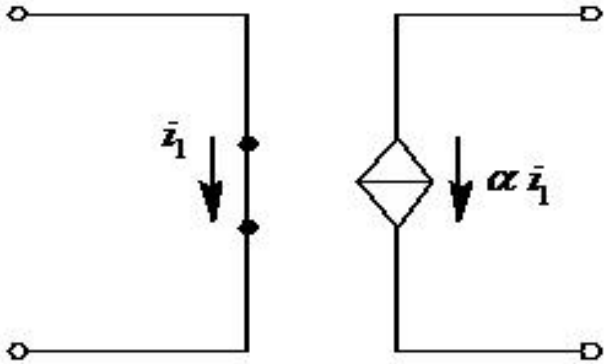
电压控制电流源VCCS



电流控制电压源CCVS



电流控制电流源CCCS

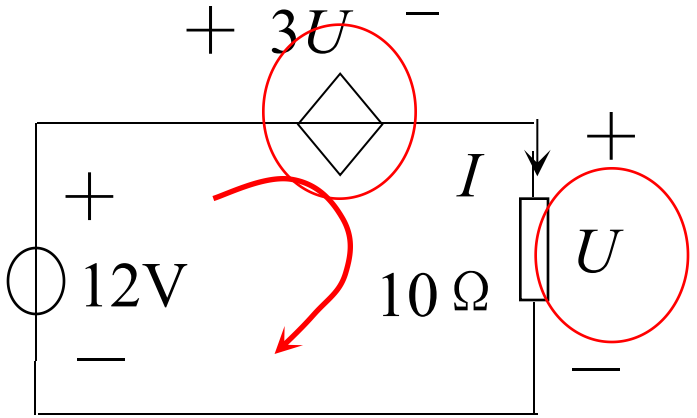


八 → 受控源（controlled source）

2 含受控源电路的分析

受控源→独立源

例 如图，求受控源吸收的功率。



解： $3U + U = 12$

$$U = 3\text{V}$$

$$I = \frac{U}{10} = 0.3\ (\text{A})$$

$$P = 3U \cdot I = 3 \times 3 \times 0.3 = 2.7\ (\text{W})$$